



Comparative evolutionary genetics of deleterious load in sorghum and maize

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Sorghum and maize share a close evolutionary history that can be explored through comparative genomics^{1,2}. To perform a large-scale comparison of the genomic variation between these two species, we analysed ~13 million variants identified from whole-genome resequencing of 499 sorghum lines together with 25 million variants previously identified in 1,218 maize lines. Deleterious mutations in both species were prevalent in pericentromeric regions, enriched in non-syntenic genes and present at low allele frequencies. A comparison of deleterious burden between sorghum and maize revealed that sorghum, in contrast to maize, departed from the domestication-cost hypothesis that predicts a higher deleterious burden among domesticates compared with wild lines. Additionally, sorghum and maize population genetic summary statistics were used to predict a gene deleterious index with an accuracy greater than 0.5. This research represents a key step towards understanding the evolutionary dynamics of deleterious variants in sorghum and provides a comparative genomics framework to start prioritizing these variants for removal through genome editing and breeding.

Sorghum (*Sorghum bicolor* L. Moench) and maize (*Zea mays* L.) are both members of the Poaceae family and often serve as a model system for comparative plant genomics. Their common Poaceae ancestor underwent a whole-genome duplication event ~96 million years ago², and a second whole-genome duplication in maize corresponds closely with its divergence 12 million years ago from sorghum¹. The role of polyploidization in maize diversification² makes the sorghum–maize system particularly powerful for comparative studies.

Archaeobotanical studies support a single sorghum domestication event around 3000 BC in eastern Sudan³, and genetic studies support a potential second independent domestication centre in west Africa^{4,5}. Maize, in contrast, was domesticated once from teosinte in the Balsas River valley in central Mexico around 9,000 years ago^{6,7}. While some orthologues between these two species experienced parallel selection during domestication⁸, most domestication-related

genes seem to be drawn from a non-overlapping set⁹. It is well established that maize experienced a decline in effective population size due to a domestication bottleneck^{10,11} that increased the burden of deleterious alleles in the domesticate compared with teosinte^{12,13}. Evidence for reduced nucleotide diversity in landraces from a genetic bottleneck⁵ or a population size decline¹⁴ has also been reported for sorghum.

Sorghum has a hermaphroditic inflorescence, contributing to its predominantly self-pollinating nature. Domesticated sorghum has an estimated outcrossing rate of only 7 to 20% (refs. ^{9,15}), while the outcrossing rate tends to be higher (up to 70%; ref. ¹⁶) for weedy and wild sorghum. In contrast, maize and its wild progenitor teosinte are monoecious, with an outcrossing rate of over 90% (ref. ¹⁷). In this study, we performed a joint analysis of functional variation in sorghum and maize lines that span the wild-to-domesticated continuum to compare deleterious mutation accumulation in these two closely related species with distinct domestication histories and mating systems.

We first conducted an extensive characterization of the levels and patterns of standing genetic variation in a diversity panel of 499 sorghum lines using whole-genome resequencing. These accessions represented a wide range of molecular and phenotypic diversity, including wild relatives, landraces and improved breeding lines (Supplementary Table 1). The average sequencing depth per line was 17× (Supplementary Fig. 1). Variants were called using the *S. bicolor* reference genome (v.3.1)¹⁸ (Supplementary Fig. 2), followed by filtering to obtain a high-quality core set of 13.2 million single nucleotide polymorphisms (SNPs) and 1.8 million indels (Supplementary Fig. 3).

Five morphological forms (botanical races) have been previously defined within *S. bicolor*: bicolor, durra, guinea, caudatum and kafir. *S. bicolor* race bicolor was the first race to be domesticated^{19,20}, while the remaining four ‘modern’ races exhibited parallel evolution for free-threshing grains. Consistent with previous reports, we observed population structure at both the race (Fig. 1a) and geographical levels using principal component analysis^{21,22} (Supplementary Fig. 4).

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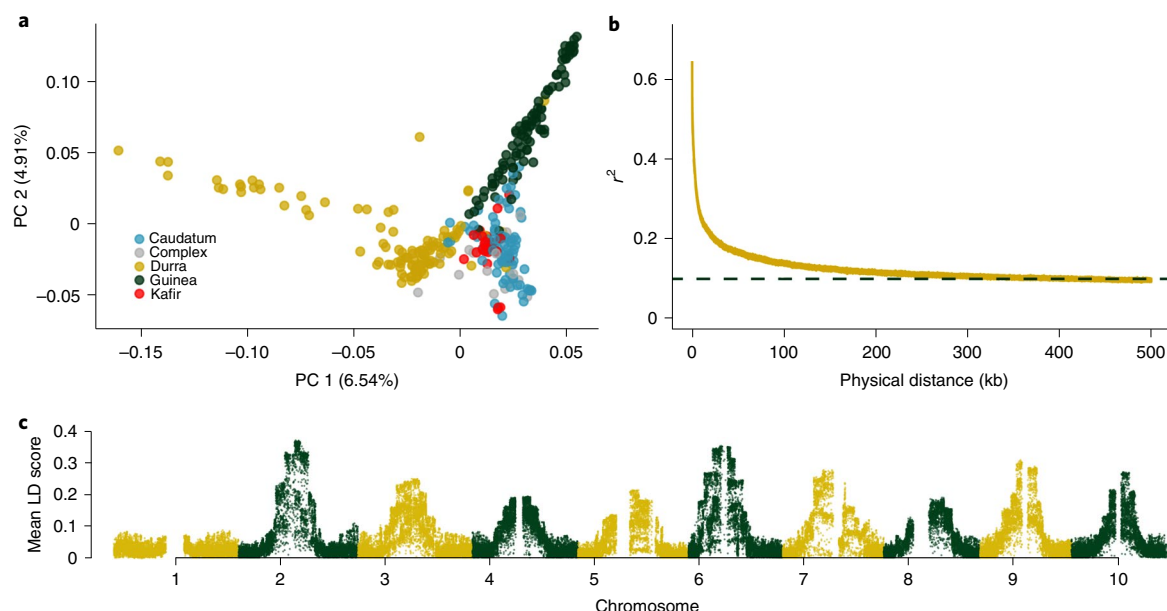


Fig. 1 | Population structure and LD patterns in sorghum. **a**, Principal component (PC) analysis of sorghum lines. **b**, LD decay rate as determined by the squared correlations of allele frequencies (r^2) against physical distance (kb) between polymorphic SNP loci. **c**, Mean LD scores estimated in 1 Mb windows. The reference genome is missing the majority of the centromeric region on chromosome 1.

Linkage disequilibrium (LD) in the diversity panel decayed rapidly to 50% of its initial value within the first few kilobases (kb), reaching background levels ($r^2=0.1$) around 300 kb (Fig. 1b). We observed slight differences in LD decay rates (Supplementary Fig. 5a) and more striking allele frequency differentiation (fixation index (F_{ST}), 0.16–0.32; Supplementary Fig. 5b,c) among races, recapitulating sorghum's domestication and adaptation history in variable environments. Local LD profiles revealed a significant increase in LD around the centromeres on all chromosomes except chromosome 1 (Fig. 1c), where a large genomic interval surrounding the centromere is missing from the reference genome²³.

To allow for a comparative analysis, genomic evolution and amino acid conservation modelling^{24,25} were used to catalogue candidate deleterious variants across both the sorghum and maize genomes. Genomic evolutionary rate profiling (GERP)²⁶ identified 64.9 megabases (Mb) (ref. ²⁵) of the sorghum genome (9.49%) as evolutionarily constrained (GERP > 0). In maize, this value increased to 117 Mb (ref. ²⁷) but represents only 4.16% of its genome. As a complement, sorting intolerant from tolerant (SIFT)²⁸ scores were calculated to predict the effect of amino acid substitutions on protein function for both maize and sorghum. In sorghum, of the 459,188 SNPs identified within exons, 19% (87,684) were considered putatively deleterious (SIFT < 0.05). Similarly, 8% (53,583) of the exonic SNPs in maize were annotated as deleterious. As expected, we found that a large percentage of the variants inside coding regions were also evolutionarily constrained (GERP > 0) in both sorghum (44%) and maize (53%) (Supplementary Fig. 6).

We split coding variants into five categories based on SIFT and GERP scores: deleterious (GERP ≥ 2, SIFT < 0.05), non-conserved deleterious (GERP < 2, SIFT < 0.05), stop mutations (either gain or loss), tolerated (non-synonymous, SIFT > 0.05) and synonymous mutations (Supplementary Fig. 7). Similar to maize²⁹, the sorghum derived allele frequency spectrum (Supplementary Fig. 8) showed that the categories of deleterious mutations exhibited an excess of low-frequency variants compared with non-deleterious variants. As in maize²⁷, deleterious mutations were also enriched in sorghum pericentromeric regions where suppressed recombination makes it difficult for the organism to purge these variants (Supplementary Fig. 9).

After maize and sorghum split from a common ancestor, the former went through a whole-genome duplication around ~12 million years ago^{1,30}. Once both maize subgenomes were combined within a single nucleus, the process of fractionation commenced, whereby one copy of each duplicated gene pair tends to be lost³⁰. We investigated whether the fractionation and syntenic status between these two species are associated with the accumulation of deleterious variants. We found the proportion of variants in each of the five categories to be similar between fractionated and non-fractionated genes in both maize and sorghum, but the number of deleterious variants was significantly higher in non-syntenic genes for both species (Fig. 2a). Our data suggest that non-syntenic genes, potentially less essential, carry deleterious variants at higher levels in both species.

We also explored the deleterious burden of wild, landrace and improved lines in both maize and sorghum. The cost-of-domestication hypothesis³¹ predicts that the process of domestication and crop improvement is likely to result in an increased number of deleterious variants in the genome. Our results in maize agreed with previous findings of an excess of overall deleterious alleles among improved maize lines compared with the wild relative *Z. mays* ssp. *parviglumis*¹² (Fig. 2b). In sorghum, however, wild relatives had the highest accumulation of deleterious alleles. When analysing each sorghum race independently, we observed three tiers of deleterious mutations, whereby wild relatives had the highest burden, followed by the bicolor race. The modern agro-climatic adapted races showed the lowest burden (Supplementary Fig. 10e). This departure could be explained partially by the inherent difference in the mating systems of sorghum (selfing) and maize (outcrossing), especially from the transition to higher selfing rates in sorghum after domestication. Notably, simulations that include an increase in the rate of selfing after domestication recapitulated the decrease in the genetic load of landraces compared with wild lines, which coheres with our interpretation of the data (Fig. 2c and Supplementary Fig. 11). This difference in burden is not driven by a lower genetic diversity observed in wild sorghum, as they follow the expected pattern (wild > landrace > improved) (Fig. 2d and Supplementary Fig. 12). A number of researchers have highlighted the complexity of the sorghum domestication process, including the potential for

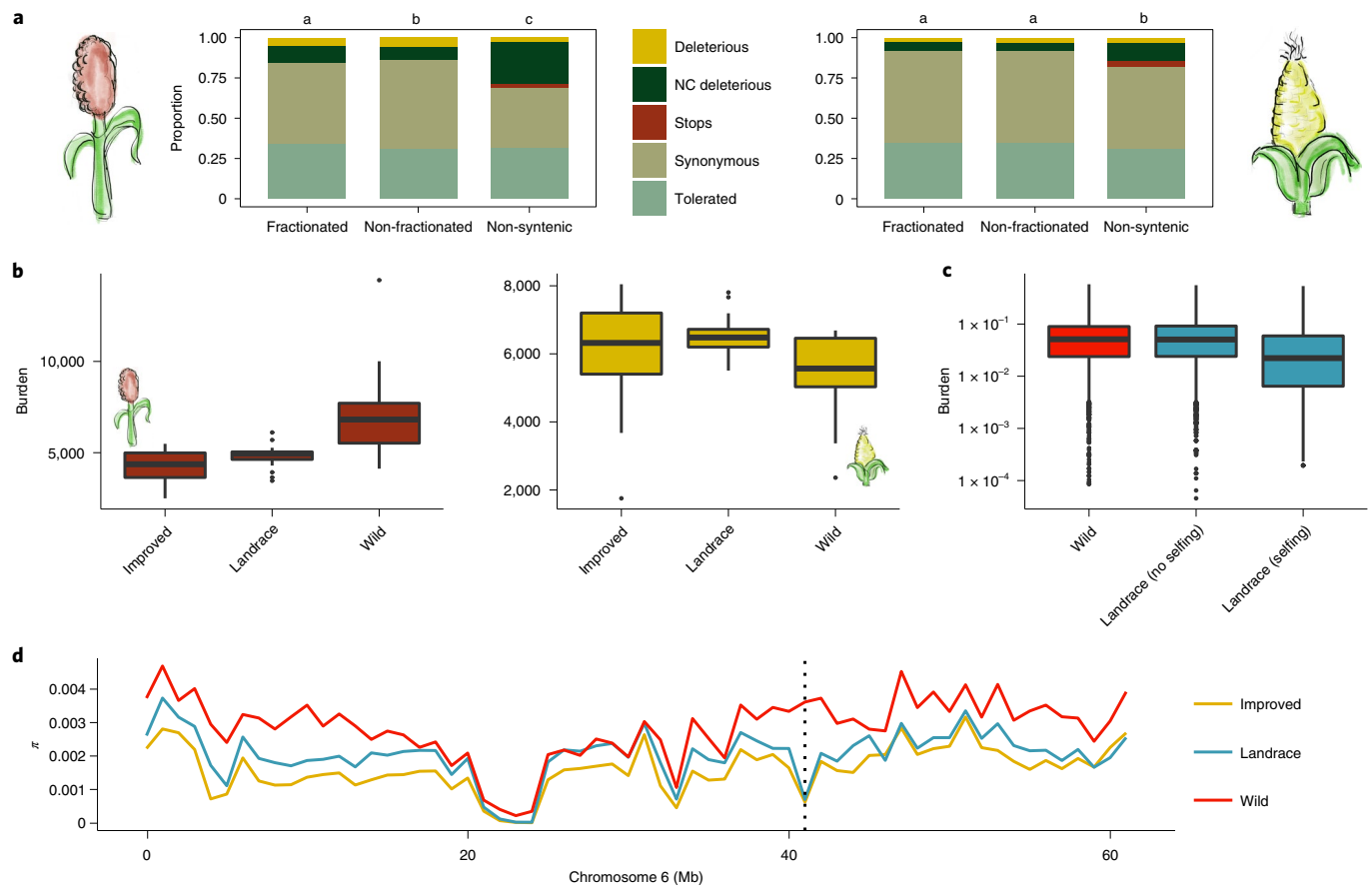


Fig. 2 | Comparative analysis of deleterious alleles in sorghum and maize. **a**, Distribution of variants inside the CDS for sorghum (left) and maize (right). The variants were divided into five categories: synonymous mutations (mutations that do not change the encoded amino acid), tolerated mutations (non-synonymous mutations, SIFT > 0.05), stop-codon mutations (either gain or loss), non-conserved (NC) deleterious mutations (SIFT < 0.05, GERP < 2) and conserved deleterious mutations (SIFT < 0.05, GERP > 2). The genes were divided into three categories: fractionated (sorghum gene with one syntologue in maize), non-fractionated (sorghum gene with two syntologues in maize) and non-syntenic. The different letters represent significant differences between groups using a two-proportion Z-test. **b**, Deleterious burden was calculated for sorghum (left) and maize (right). The additive burden was calculated as the sum of derived alleles at deleterious sites. Normalized burden calculations, as well as recessive and dominant burden, are also shown in Supplementary Fig. 10a–d. Sample sizes for sorghum: improved, $n=15$; landrace, $n=18$; wild, $n=20$. Sample sizes for maize: improved, $n=49$; landrace, $n=25$; wild, $n=20$. **c**, Distribution of genetic load under a simulation scenario. The simulations were based on the mean population size 10,000 years ago for sorghum landrace and wild lines inferred by SMC++ (ref. ³¹). The effect of increased selfing during sorghum domestication was explored by changing the inbreeding probability from 0 (no selfing) to 1 (selfing). For each box plot, $n=4,059$. The boxes span from the first to the third quartiles, the centre lines represent the median values and the whiskers show the data that lie within the 1.5 interquartile range of the lower and upper quartiles. The data points at the ends of the whiskers represent the outliers. **d**, Nucleotide diversity (π) estimates across chromosome 6 were calculated in 1Mb bins for improved (yellow), landrace (blue) and wild lines (red). The dotted line marks a major decline in nucleotide diversity for improved and landrace lines relative to wild lines. This genomic region (42 Mb) colocalizes with *dw2* and *ma1*, which are known height and maturity loci²¹. See Supplementary Fig. 12 for plots of all ten chromosomes.

introgression from wild relatives to have reduced the genetic load in modern lines^{3,21,22,32}. Simple tree-based tests confirm evidence of gene flow into some landraces (Supplementary Table 2); however, the total genetic burden in these lines is no lower than in landraces lacking evidence of introgression (Supplementary Fig. 13). While the complex domestication history of sorghum may have also contributed to the differences in load observed between wild and cultivated sorghum, our simulations suggest that the mating system probably played a prominent role. We also observed evidence for hitchhiking of deleterious variants at sites with positive selection. Specifically, when comparing the windows with the strongest evidence for positive selection (integrated haplotype score (iHS), top 5%) with those having the lowest 5%, we observed that the former had a significantly higher ratio of deleterious/non-deleterious variants than the latter (0.48 versus 0.36; $P=0.0338$; Mann–Whitney U -test). The top 5% value was still higher than the global average

(0.43), although that difference was not statistically significant at $\alpha=0.05$.

Given the recent development of supervised machine learning applied to population genetics and genomics inference^{33,34}, we evaluated the efficacy of convolutional neural networks (CNNs) in building an evolutionary model capable of predicting a deleterious index (average SIFT score) and syntenic state for sorghum genes. We split the sorghum genome into segments with each gene as a centroid and calculated 12 features per window, four of which were derived from maize. We incorporated predictors of functional importance, conservation of synteny and fractionation of maize genes, levels of gene expression variance, and several molecular evolution statistics (Fig. 3).

Through the implementation of this model, we predicted the average SIFT score per gene, a statistic that reflects the likelihood that a genic mutation would be deleterious. Our model had a prediction

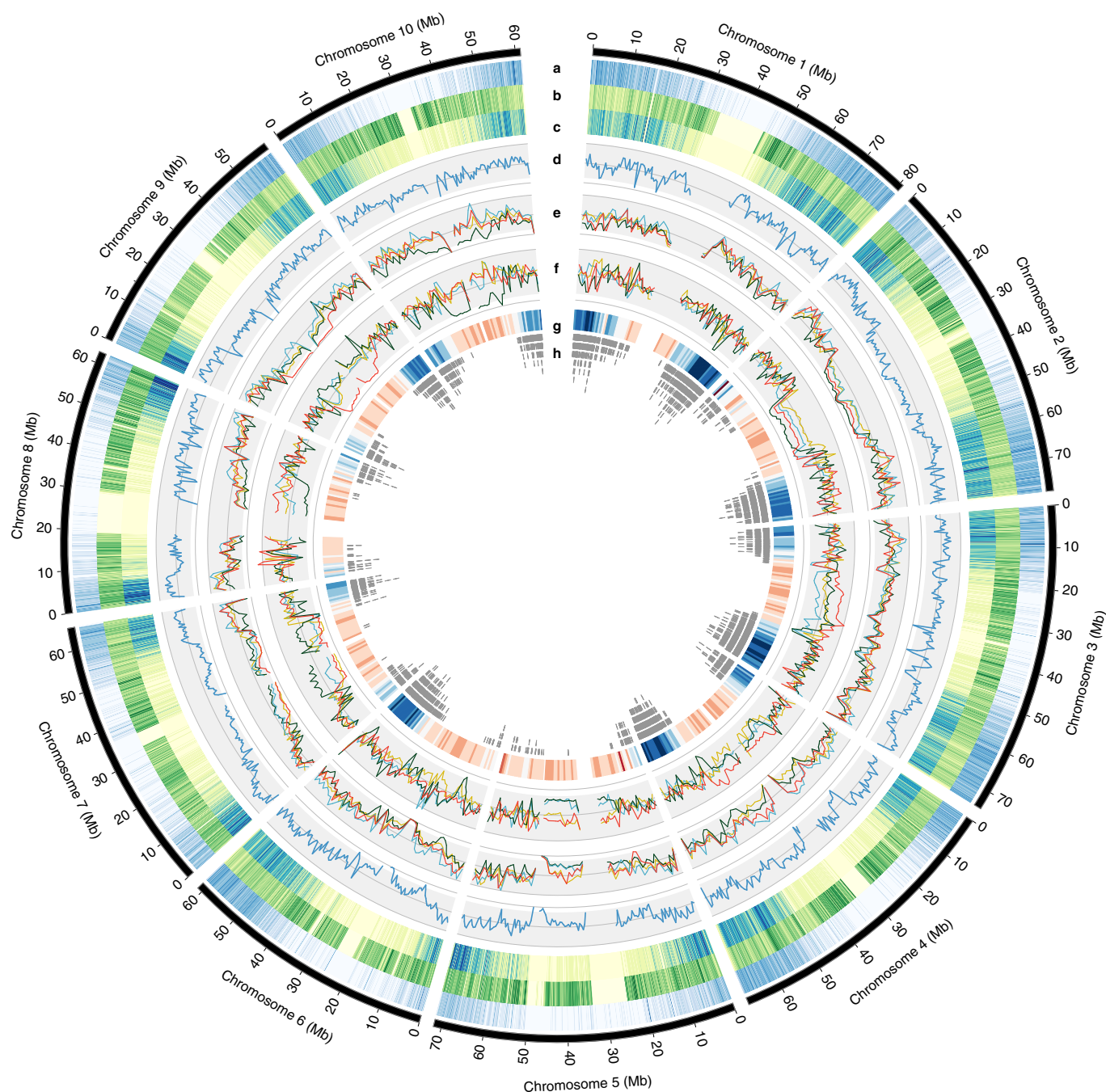


Fig. 3 | Genomic landscape of sorghum. **a**, Gene density heat map. **b**, SNP density. **c**, Indel density. **d**, Population recombination rates (ρ) were calculated using FASTPRR in 500 kb windows. **e**, Nucleotide diversity (π) in 1 Mb windows for each sorghum race; the colour codes are the same as used in Fig. 1a (red, kafir; blue, caudatum; yellow, durra; dark green, guinea). **f**, Tajima's D for each sorghum race. **g**, Average GERP score per 1 Mb window, with a divergent red-blue scale where blue represents high GERP scores. **h**, Genomic distribution of sorghum non-fractionated genes.

accuracy of 0.53 and outperformed linear regression models by 10% (Supplementary Fig. 14). The importance of individual features was assessed with a leave-one-variable-out approach. We found that four features were the most impactful. Two of them—average GERP score and number of variants in the coding sequence (CDS)—were expected to be important, as they both reflect the strength of purifying selection at a locus (Fig. 4a,b). The other two impactful features were RNA expression variance in sorghum and maize syntelogs, supporting previous research in maize where the dysregulation of expression was correlated with rare-allele burden³⁵. We also used the model to predict the syntenic state of the

focal gene in each window with an area under the curve (AUC) of ~ 0.9 . Two maize-related features were most relevant: ssw, a measure of conservation between two genomes using short k -mer alignments (Methods), and maize nucleotide diversity (π) (Fig. 4c). We expect maize syntenic genomic regions (either fractionated or non-fractionated) to have lower nucleotide diversity when compared with non-syntenic genes that are potentially non-essential. Nucleotide diversity may therefore be serving as a clear indicator of which genes are syntenic or non-syntenic, while the other model features might determine whether genes are fractionated or non-fractionated.

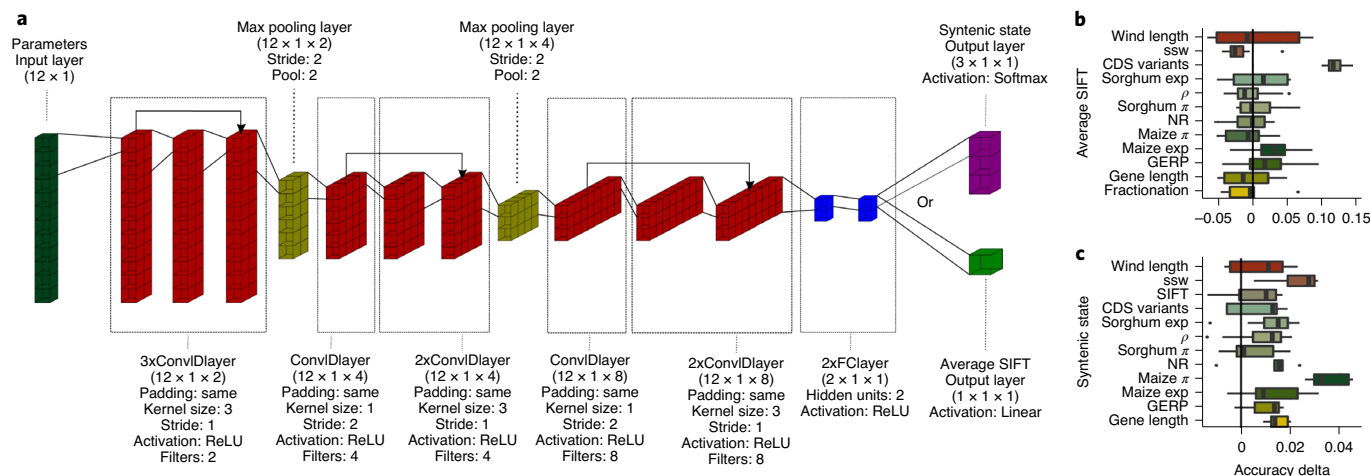


Fig. 4 | CNN architecture. **a**, The sorghum genome was divided into 34,028 windows, with each gene serving as a centroid. For each window, 12 features were calculated. Four of these features were calculated to exploit the syntenic relationships between maize and sorghum: nucleotide diversity of the syntenic segment in maize (maize π), ssw short alignments (Methods) between sorghum and maize, coefficient of variation of the expression from syntenic maize genes (maize exp), and syntenic state (fractionation). The remaining features were calculated in sorghum: nucleotide diversity (sorghum π), conservation score (mean GERP score per window, GERP), neutral substitution rate (NR) per focal gene, average SIFT score (SIFT), number of variants in the CDS of the focal gene (CDS variants), population recombination rate (ρ), coefficient of variation of the expression of the focal gene across different developmental stages and treatments (sorghum exp), window length (wind length), and focal gene length (gene length). The CNN was trained to predict the average SIFT score per gene and the syntenic state of the focal gene (fractionated, non-fractionated or non-syntenic). **b,c**, To assess the relative importance of each feature, the decrease in accuracy (accuracy delta) was calculated after removing a single feature for predicting the average SIFT score (continuous) (**b**) and syntenic state (categorical) (**c**). For **b** and **c**, the distribution in the box plots represents tenfold cross-validation ($n=10$). The boxes span from the first to the third quartiles, the centre lines represent the median values and the whiskers show the data that lie within the 1.5 interquartile range of the lower and upper quartiles. The data points at the ends of the whiskers represent the outliers.

In summary, we performed a joint variant analysis of sorghum and maize genomic variation across the wild-to-domesticated continuum to highlight their differences in genetic load accumulation. We also constructed and tested an evolutionary model using supervised machine learning that exploited the syntenic relationship between these two species. We envision that similar strategies^{36,37} could improve functional annotations and predictions across the genome of relatively less characterized species, with deep learning models initially trained on evolutionarily related model species that have extensive experimentally validated data (for example, methylation, histone marks and ATAC-seq). Overall, we constructed a genomic landscape of the sorghum genome that can be used to support genome-wide association study (GWAS) results, support variance component estimates, and inform whole-genome prediction in a comparative genomics framework.

Methods

Plant material. The plant material used in this study consists of sorghum lines that represent a wide range of molecular and phenotypic diversity. Overall, 499 lines (482 of which are unique) were used to build a sorghum haplotype map. Only unique lines were considered for further analysis. The material has three different origins: Mace et al.³ ($n=42$), the TERRA-MEPP project ($n=254$, <https://terra-mepp.illinois.edu>) and TERRA-REF ($n=203$, <http://terraref.org/>). Information for each line is included in Supplementary Table 1.

Sequencing and variant identification. The samples from all three datasets were sequenced using Illumina paired-end reads of 100 base pairs (bp) or 150 bp in length. The sequencing depth for each line ranged from 0.03 to 116.06 $\times$ (Supplementary Table 1 and Supplementary Fig. 1). This was estimated on the basis of the number of reads, the expected sorghum genome size (732 Mb) and the read length. Variant calling was performed using Sentieon's DNaseq pipeline 201711.03 (ref. ³⁸), a proprietary implementation of the GATK variant calling pipeline³⁹. Briefly, BWA⁴⁰ v.0.7.13 was used to align the raw sequence reads to the BTx623 reference genome v3.1 available in Phytozome¹⁸. Next, duplicated reads were removed, and local realignment was performed around the indels. Base quality score recalibration was then performed using SNP markers identified in a previous sorghum hapmap²⁵ built with 240 TERRA-MEPP accessions.

Finally, gvcf files were generated using the Haplotyper in the emit_mode gvcf. The 485 gvcfs were then jointly called to produce a single VCF file using the GVCFTyper mode. For more details on the pipeline, see Supplementary Fig. 2a and the code in the GitHub repository (<https://github.com/GoreLab/Sorghum-HapMap>).

After variant calling, a vcf file containing ~41 million markers was generated. Of these, 35,025,902 were biallelic SNPs, and 3,486,787 were biallelic indels (see Supplementary Fig. 2b for the details). Hard filters were applied using GATK best practices recommendations. Briefly, SNP markers having a QualByDepth < 2, FisherStrand > 60.0, RMSMappingQuality < 40.0, MappingQualityRankSumTest < -12.5 or ReadPosRankSumTest < -8.0 were removed. Similarly, indels with QualByDepth < 2.0, FisherStrand > 200.0 or ReadPosRankSumTest < -20.0 were also removed (Supplementary Fig. 2b). The average inbreeding coefficient at filtering STAGE 2 was 0.68. This value contrasted with our expectation of high homozygosity for sorghum inbred lines. We applied to the data an allele balance filter (see the GitHub repository [vcfaddanot.py](https://github.com/GoreLab/vcfaddanot.py)) that calculated the ratio between the number of reads supporting the heterozygous (het) calls and the total number of reads:

$$AB = \frac{\text{number of reads supporting het call}}{\text{total number of reads}} \times 100$$

According to previous research⁴¹, most of the variants with an AB value above 30 could be validated using Sanger sequencing. Applying this same threshold, heterozygous calls with an AB value lower than 30 were masked. Additionally, vcfutils⁴² was used to mask genotypes with a read depth (minDP) lower than 4, markers that were monomorphic and markers with call rates lower than 50%. Finally, SNPs with an inbreeding coefficient lower than 0 were removed. The inbreeding coefficient per marker was calculated as:

$$IC = 1 - \frac{H_{obs}}{H_{exp}}$$

where H_{obs} and H_{exp} are the observed and expected heterozygosity under Hardy-Weinberg equilibrium. Finally, missing genotypes were imputed and phased into haplotypes using Beagle v.4.1 (ref. ⁴³) using a default window size of 50,000 SNPs and an $N_e=150,000$. In total, 13,170,712 SNPs and 1,804,397 indels were called and imputed for the 485 accessions. After these SNPs and indels were identified, we included sequences for 14 *S. bicolor* subsp. *verticilliflorum*. Similarly, gVCFs were created, and a joint calling procedure was performed using the entire dataset. The resulting VCF was filtered down to the 13 million SNP positions, and the same filters were applied (Supplementary Fig. 2c). Variants were called for all lines

($n = 499$); however, only unique lines with sequencing depth higher than $2\times$ were considered for further analysis ($n = 480$).

LD. LD decay was calculated using PopLDdecay v.3.31 (<https://github.com/BGI-shenzhen/PopLDdecay>). Measures of LD (r^2) were calculated for the entire population and for each chromosome and subpopulation. Pairwise r^2 estimates were calculated from the unimputed SNP dataset with MAF > 0.05 and maximum missing rate < 0.25 . LD scores were calculated using the Genome-wide Complex Trait Analysis suite^{44,45} with the default settings.

Deleterious mutations. *Candidate deleterious alleles identification.* We used SIFT (ref. ⁴⁶) to annotate the sorghum and maize SNP data. The SIFT algorithm predicts whether an amino acid substitution is deleterious to protein function⁴⁸. SIFT uses protein alignments to identify conserved amino acids and provide a score of the putative deleterious effect for each position of the protein. These scores range from 0 to 1, and positions with a SIFT score < 0.05 are predicted to be deleterious.

In sorghum, GERP scores²⁶ estimated from the alignment of six species (*Z. mays*, *Oryza sativa*, *Setaria italica*, *Brachypodium distachyon*, *Hordeum vulgare* and *Musa acuminata*)⁴⁷ were used to identify evolutionarily constrained nucleotides. From the ~13 million SNPs, 455,546 variants were located within the CDS and were split into five categories based on SIFT and GERP scores as follows: synonymous mutations (mutations that do not change the encoded amino acid), tolerated mutations (non-synonymous mutations, SIFT > 0.05), stop-codon mutations (either gain or loss), non-conserved deleterious mutations (SIFT < 0.05 , GERP < 2) and conserved deleterious mutations (SIFT < 0.05 , GERP > 2).

The derived site frequency spectrum was calculated for the 455,546 CDS variants using both *Z. mays* (AGPv.3.22 annotation)⁴⁸ and *Setaria italica* (*Setaria italica* v2.2, DOE-JGI, <http://phytozome.jgi.doe.gov/>) as outgroups to determine the ancestral state. Briefly, each sorghum gene was aligned with its syntenic orthologues⁴⁹ using Clustal Omega⁵⁰ in a two-step approach. We first used Clustal Omega to align the protein sequences, and only then we used tranalign (EMBOSS) to align the CDS for both maize and setaria. For each variant, the corresponding nucleotides in maize (both subgenomes) and setaria were identified, and est-sfs software⁵¹ was used to infer the probability of the derived versus ancestral allelic state. The same approach was used to annotate deleterious alleles in maize. Briefly, maize HapMap v.3 (ref. ⁵²) (1,200 individuals) was used to annotate deleterious alleles. Only markers with the LLD flag present and the N15 flag absent were used, as suggested by the authors. The LLD flag includes SNPs that are confirmed to be in LD with genotyping-by-sequencing (GBS) anchor markers, and the N15 flag is used to mark SNPs within 5 bp of an indel. Together, the LLD flag present and the N15 flag absent represent the cleanest marker set⁵². Maize annotation on the v.3 reference genome (AGPv.3.22)⁴⁸ was used to calculate SIFT scores. GERP scores were available from Rodgers-Melnick et al.⁵³. Sorghum and setaria were used as outgroups for maize to infer the derived/ancestral allele using sts-ufs software⁵¹.

Using a single reference genome for a species when calculating GERP and SIFT scores introduces a bias that underestimates the number of deleterious alleles in domesticated material versus wild relatives¹². It has been previously shown that the annotation of markers for which the reference allele is derived might be unreliable⁵⁴. To mitigate this bias, we calculated the probability that each reference-derived allele would have been classified as it was (deleterious, non-conserved deleterious and so on) had the reference allele been ancestral separately for maize and sorghum. Briefly, all the alleles for which BTx623 (the sorghum reference genome genotype) or B73 (the maize reference genome genotype) was ancestral were divided into bins of 1% derived allele frequency, and the fraction of reference-ancestral sites in each functional category was calculated. Those fractions were used as weights for all the reference-derived sites.

Deleterious burden calculations. The fitness for a variant was calculated relative to the ancestral state⁵⁵. The ancestral allele was set as the non-deleterious marker (coded as 0 in the dosage matrix), and the derived allele as the potential deleterious allele. We then calculated the burden of each line using the unimputed marker matrix. Different ways to calculate the deleterious burden were defined. Under a dominant model, we counted each site with at least one deleterious allele. Under the additive model, we counted the total number of alleles that were alternative to the ancestor. Here each genotype homozygous for the deleterious allele would score 2, while a heterozygous genotype scored 1. In the recessive model, only the sites that were homozygous for potentially deleterious variants were counted. Finally, under the normalized additive model, we calculated the additive burden and divided it by the number of available deleterious loci. The idea behind this last statistic is that it might control for individuals with higher missing call rates. The individuals used for the comparison between improved lines, landraces and wild relatives for maize and sorghum are included in Supplementary Table 3.

Deleterious burden simulations. To understand the potential reason for decreased burden in landraces compared with wild sorghum, we simulated different scenarios. To parameterize our simulations to fit the empirical data, we estimated the population history of sorghum using SMC++ (ref. ⁵⁶). We selected unambiguous individuals from each group (19 wild, 18 landrace and 14 improved) to ensure equal sampling. We used the repeat masking of the sorghum reference

genome v.3.0.1 to create input files for each line of a group as a distinct lineage. We used forward-in-time simulations in fwdpy11 (<https://github.com/molpopgen/fwdpy11>), a Python package using the fwdpp library⁵⁷, to simulate a diploid random-mating population. We simulated a 100Mb region where deleterious mutations occurred at 1% of a neutral mutation rate of 3×10^{-8} mutations per bp per generation and used a recombination rate equal to the neutral mutation rate.

The effect size of deleterious mutations followed an exponential distribution with a mean of -0.05 . On the basis of the mean population size 10,000 years ago for the landrace and wild lines inferred by SMC++, we simulated an equilibrium population having a constant size of $N_{anc} = 26,696$ for nine N_{anc} generations before changing parameters. The demographics used followed the inferred demography for the past 10,000 years. We tested the effect of increased inbreeding from selfing during sorghum domestication by changing the inbreeding probability from 0 (complete outcrossing) to 1 (complete inbreeding). We replicated each parameter combination 100 times. At the end of the simulation, we recorded the fitness of 30 random individuals. We calculated burden as the difference in fitness of an individual compared with the fittest individual across all simulations.

Evidence of positive selection, hitchhiking test and gene flow test. We calculated the iHS on the three sorghum groups (improved, landrace and wild) independently using vcfliib⁵⁸. We ran the test using overlapping windows and discarded values that failed integration. Each of the features used in the model are detailed in the next section. We used an approach similar to Zhang et al.⁵⁹ to assess whether the deleterious patterns observed in sorghum were also influenced by hitchhiking (the enrichment of deleterious mutations in genomic regions with evidence of positive selection). Briefly, normalized |iHS| scores were averaged in 100kb bins across the ten chromosomes in improved sorghum. Similarly, the proportion of deleterious to non-deleterious variants in improved sorghum was also averaged in 100kb windows across the ten chromosomes. The ratio of deleterious to non-deleterious variants was compared between windows with high and low iHS using a Mann–Whitney *U*-test.

The *D* statistic (ABBA-BABA) was calculated using ANGSD software⁶⁰. The lines used to calculate Patterson's *D* statistic are specified in Supplementary Table 2. We calculated the *D* statistic using the durra and caudatum landraces as a focal group. *Sorghum halepense* was used as an outgroup for allele polarization.

CNN model. The sorghum genome was divided into windows each containing a single focal gene. Several summary statistics from both sorghum and maize were then calculated for each window. These windows were randomly split into training, validation and test sets, and a CNN framework was used to build a model for predicting the average deleterious score (SIFT) and syntenic state. Syntenic state refers to whether the focal sorghum gene in each window is fractionated, non-fractionated or non-syntenic when compared with maize.

Defining sorghum genome windows. The midpoint distance between adjacent genes was calculated, allowing for the calculation of a series of intervals that covered each chromosome from start to end. In total, the sorghum genome was divided into 34,028 windows (Supplementary Table 4).

Nucleotide diversity (π), Tajima's *D* and per-site F_{ST} . Nucleotide diversity⁶¹ and Tajima's *D* were calculated with the sorghum and maize SNPs using vcftools site-pi⁴², TajimaD and weir-fst-pop. Using bedtools, we calculated the average $\pi/D/F_{ST}$ value per window, counting every nucleotide site inside the window and giving a value of 0 for monomorphic positions.

GERP. Sorghum GERP²⁶ scores were estimated from the alignment of six species (*Z. mays*, *O. sativa*, *Setaria italica*, *B. distachyon*, *H. vulgare* and *M. acuminata*)⁴⁷. For further information, please refer to Valluru et al.²⁵. We used maize GERP scores calculated previously by Rodgers-Melnick et al.⁵³. The average GERP scores were calculated per window. Additionally, the average neutral substitution rates (the sum of the tree branch lengths) were calculated for each focal gene.

SIFT. Sorghum SIFT annotations were calculated with a sorghum database created using the Sbicolor_313.v3.1 gene annotation. Only primary transcripts were considered. Sift4g was used to evaluate the ~13 million sorghum variants. Maize SIFT scores were calculated on a subset of the HapMap v.3 markers⁵², including only those with the LLD flag present and the N15 flag absent (29 million variants). For maize, the pre-computed SIFT database on the v.3 reference genome (AGPv3.22) was used. The average SIFT scores per gene were calculated.

SSW alignment between sorghum and maize. The sorghum reference genome was aligned to the maize genome with a sliding window of 20 bp using the Complete Striped Smith Waterman library (<https://github.com/mengyao/Complete-Striped-Smith-Waterman-Library>) for faster alignments. We reported the number of times each 20-bp tag from sorghum aligned to maize, the maximum alignment score of the 20-bp tag and the average of multiple alignment scores of each 20-bp tag. For each window used in the evolutionary model, the average maximum alignment score was calculated and used as a proxy of local conservation between the sorghum and maize genomes.

Population recombination rate (ρ). Population recombination rates were calculated in the TERRA-MEPP P1-P4 dataset (because of computational constraints) using the software FASTEPRR⁶². We also calculated the recombination rate for each of the gene windows used to build the evolutionary model (FASTEPRR_segments.R).

Sorghum and maize gene expression. Six sorghum (PRE0023, PRE0028, PRE0146, PRE0295, PRE1441 and Grass1) and two maize (B73 and Mo17) lines were grown under the same greenhouse experimental conditions. Plant material for RNA sequencing was collected from two tissues (growing point and leaves) at four developmental stages (three-, five-, six- and seven-leaf stages) during day and night conditions. In total, 192 and 64 sample combinations of genotypes, developmental stages, tissue types and diurnal conditions for sorghum and maize were obtained, respectively. The samples were processed according to Kremling et al.³⁵. Briefly, RNA was extracted using TRIzol (Invitrogen) with Direct-zol columns (Zymo Research), and 3' RNA-seq libraries were prepared from 500 ng of total RNA in 96-well plates on an NXP liquid handler (Beckman Coulter) using QuantSeq FWD kits (Lexogen) according to the manufacturer's instructions. The libraries were pooled to 96-plex and were sequenced with 90-bp single-end reads using Illumina TruSeq primers on an Illumina NextSeq 500 with v2 chemistry at the Cornell University Biotechnology Resource Center.

The first 12 bp and Illumina TruSeq adapter remnants were removed from each read using Trimmomatic v.0.32. The splice-aware STAR aligner v.2.4.2a was used to align reads against either the sorghum v.3.1 or the maize AGPv3 reference genome annotations, allowing reads to map to at most ten locations (outFilterMultimapNmax, 10) with at most 4% mismatches (outFilterMismatchNoverLmax, 0.04), while filtering out all non-canonical intron motifs (outFilterIntronMotifs, RemoveNoncanonicalUnannotated). The default settings from the STAR v.2.4.2a aligner were used to obtain gene-level counts (quantModel, GeneCounts) from the resulting BAM files.

CNN architecture. A deep residual CNN was developed (DeepEvolution), tailored to predict different population genetics parameters in classification and regression problems. The DeepEvolution architecture is composed of one input layer, nine one-dimensional convolutional layers, two max pooling layers, two fully connected layers and one output layer (Fig. 4a). To better exploit deeper low- and high-level representations from the input space, residual blocks over the network were used as motivated by the ResNet architecture⁶³ and batch normalization after convolutional layers. ReLU activations were applied on the hidden layers to better leverage nonlinearities from the population genetics parameter space. A decreasing number of channels and filter sizes were chosen, which is common in many efficient computer vision algorithms, such as the AlexNet architecture⁶⁴. For the regression problem of predicting average SIFT score, a linear activation function was used in the output layer. In the classification framework, the softmax activation function was implemented to predict the probability of the three classes of syntenic state (two copies, non-fractionated; one copy, fractionated; zero copies, non-syntenic). The network for average SIFT score was optimized using the Adam algorithm⁶⁵ with a learning rate of 0.001 to minimize the mean-squared-error function for regression, while for syntenic state, categorical cross-entropy was used for classification. Additional details about the model architecture, cross-validation and feature importance analysis can be reviewed in the documented code (<https://github.com/GoreLab/Sorghum-HapMap/tree/master/CNN/codes>). The DeepEvolution model was fitted using the library keras 2.1.6 and python 3.6. To compare with accuracy obtained from the CNN, we used linear regression (average SIFT score) and multinomial logistic regression (syntenic state) models using the same predictors as in the CNN.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The raw sequencing data for the TERRA-MEPP lines are available through the NCBI BioProject PRJNA513297. The raw data for Mace et al.³ are available through the BioProject PRJNA182489. The TERRA-REF raw data are available through the data commons database at CyVerse: <http://datacommons.cyverse.org/browse/iplant/home/shared/terraref>. The gene expression raw data are available through the BioProject PRJNA503076. The SIFT raw results and VCF files, among others, are available through the CyVerse repository: (http://datacommons.cyverse.org/browse/iplant/home/shared/GoreLab-dataFromPubs/Lozano_MaizeSorghum_2019).

Code availability

The code used throughout the article is available at the GitHub repository: <https://github.com/GoreLab/Sorghum-HapMap>

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Author contributions

R.L., E.G., E.S.B., J.R.-I. and M.A.G. designed the project and all the experiments. R.L. and J.P.R.d.S. performed the CNN analysis. R.L. and E.G. performed the bioinformatic analysis. M.G.S. performed the burden simulation analysis. R.V. and N.B. performed the transcriptome analysis. S.B.F., N.S., T.C.M., E.A.C. and P.J.B. contributed the plant material. M.T.P. constructed the genomic libraries for sequencing the 14 *S. bicolor* subsp. *verticilliflorum* accessions. P.J.B., N.S., T.C.M., E.S.B., J.R.-I. and M.A.G. contributed the whole-genome sequence data. R.L., E.G., M.G.S., J.P.R.d.S., R.V., S.B.F., J.R.-I. and M.A.G. analysed the data. R.L. prepared the figures and together with M.A.G. and J.R.-I. wrote the manuscript. All authors provided critical insights and read and approved the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Software and code

Policy information about [availability of computer code](#)

Data collection No software was used to collect data

Data analysis Software used for variant calling and variant analysis: Trimmomatic (v0.32), BWA-MEM (0.7.13), bedtools (V2.25), Sentieon (201711.03), vcftools (0.1.16), GATK (v3.7), PopLDdecay v3.31, GCTA (1.25.2), Beagle (v5), PopLDdecay (v3.31), clustal Omega (v1.2.4)
For deleterious variant analysis: SIFT 4G, GERP++ (v20110522), est-usfs (2.03), vcflib (v1.0.0-rc2), ANGSD (v0.931)
For simulations: SMC++ (v1.15.4), fwdpy11 (<https://github.com/molpopgen/fwdpy11>)
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through the CyVerse repository: (http://datacommons.cyverse.org/browse/iplant/home/shared/GoreLab/dataFromPubs/Lozano_MaizeSorghum_2019). Custom scripts used throughout this manuscript are included in: <https://github.com/GoreLab/Sorghum-HapMapci>

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Study description	<i>This study analyzed the whole genome re-sequencing data of 482 unique sorghum lines (499 in total) and identified 13 million SNPs</i>
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Study description	across the sorghum genome. Together with 25 million variants previously identified in 1,218 maize lines, we annotated deleterious variants in both species and compared their distribution and accumulation patterns across the domestication continuum. Additionally, sorghum and maize population genetic summary statistics were used to predict a gene deleterious index.
Research sample	Sorghum: The research sample utilized a set of 482 unique sorghum lines (499 in total). The unique lines were considered for further analysis. The material has three different origins: Mace et al. (https://www.nature.com/articles/ncomms3320) (n = 42), the TERRA-MEPP project (n = 240, https://terra-mepp.illinois.edu), and TERRA-REF (n = 203, http://terraref.org/). Mace et al. dataset was meant to be a first generation HapMap of Sorghum, lines were selected to maximize genetic diversity and to include wild relatives, landraces and members of each of the sorghum botanical races. TERRA-MEPP lines were in its majority biomass sorghum and TERRA-REF varieties included bot a diversity subset and biomass sorghum lines. By combining these dataset we wanted to cover as much of the genetic diversity in sorghum as possible. Maize: We used the variants discovered in a diversity panel of 1,218 maize lines previously described in the third-generation Zea mays haplotype map paper by Bukowski et al. (2018)
Sampling strategy	No preliminary analysis on sample size were performed. To maximize the number of markers that we discovered we used all the sequencing data available. The latest study (Mace et al. Nature Communications 2013) used WGS on only 42 sorghum lines. We have significantly increased this number to 482 unique sorghum lines.
Data collection	Sorghum lines under the TERRA-MEPP project were grown in a greenhouse, genomic DNA libraries were prepared locally, and sequencing was performed by Novogene except for 14 lines (P5) which were sequence at the UC Davis sequencing center. All the data was submitted to Cornell University through an FTP connection and stored in the Gore Lab server by Roberto Lozano.
Timing and spatial scale	TERRA-MEPP libraries were prepared in 2017 and DNA sequencing occurred between 2017 and 2018. The last batch of sequencing was prepared and sequenced in 2020.
Data exclusions	We did not consider replicated sorghum lines in the analysis. Only for quality control. This exclusion criteria was pre-established to avoid bias in the deleterious analysis due to duplicated samples.
Reproducibility	For DNA sequencing, we used biological replicates. Consistent markers between the same line sequenced by different groups (TERRA-MEPP vs TERRA-REF) represented a quality indicator. All attempts at replication were successful (assessed by clustering of the genomic relationship matrices). To maximize reproducibility in the data analysis, we have included a publicly available code repository that can be used to replicate the results presented here. The results in this study were reproduced internally using the available code and all attempts to repeat the experiments were successful.
Randomization	No randomization was needed for performing WGS. Samples (or sites) were randomly chosen in related experiments.
Blinding	The variant calling procedure, deleterious mutation identification and burden calculations are inherently blinded procedures. Samples were processed identically through standard and in some cases automated procedures (DNA sequencing, Next Generation Sequencing, genotype calling, deleterious mutation annotation) that should not have bias outcomes.
Did the study involve field work?	☐ Yes ☒ No

Field work, collection and transport

Field conditions	<i>Describe the study conditions for field work, providing relevant parameters (e.g. temperature, rainfall).</i>
Location	<i>State the location of the sampling or experiment, providing relevant parameters (e.g. latitude and longitude, elevation, water depth).</i>
Access & import/export	<i>Describe the efforts you have made to access habitats and to collect and import/export your samples in a responsible manner and in compliance with local, national and international laws, noting any permits that were obtained (give the name of the issuing authority, the date of issue, and any identifying information).</i>
Disturbance	<i>Describe any disturbance caused by the study and how it was minimized.</i>

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	<i>Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.</i>
Validation	<i>Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.</i>

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)	<i>State the source of each cell line used.</i>
Authentication	<i>Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.</i>
Mycoplasma contamination	<i>Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.</i>
Commonly misidentified lines (See ICLAC register)	<i>Name any commonly misidentified cell lines used in the study and provide a rationale for their use.</i>

Palaeontology and Archaeology

Specimen provenance	<i>Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the issuing authority, the date of issue, and any identifying information).</i>
Specimen deposition	<i>Indicate where the specimens have been deposited to permit free access by other researchers.</i>
Dating methods	<i>If new dates are provided, describe how they were obtained (e.g. collection, storage, sample pretreatment and measurement), where they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.</i>
☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.	
Ethics oversight	<i>Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.</i>

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	<i>For laboratory animals, report species, strain, sex and age OR state that the study did not involve laboratory animals.</i>
Wild animals	<i>Provide details on animals observed in or captured in the field; report species, sex and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.</i>
Field-collected samples	<i>For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.</i>
Ethics oversight	<i>Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.</i>

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Human research participants

Policy information about [studies involving human research participants](#)

Population characteristics

Describe the covariate-relevant population characteristics of the human research participants (e.g. age, gender, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."

Recruitment

Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.

Ethics oversight

Identify the organization(s) that approved the study protocol.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration

Provide the trial registration number from ClinicalTrials.gov or an equivalent agency.

Study protocol

Note where the full trial protocol can be accessed OR if not available, explain why.

Data collection

Describe the settings and locales of data collection, noting the time periods of recruitment and data collection.

Outcomes

Describe how you pre-defined primary and secondary outcome measures and how you assessed these measures.

Dual use research of concern

Policy information about [dual use research of concern](#)

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

- | No | Yes |
|--------------------------|---|
| ☐ | ☐ Public health |
| ☐ | ☐ National security |
| ☐ | ☐ Crops and/or livestock |
| ☐ | ☐ Ecosystems |
| ☐ | ☐ Any other significant area |

Experiments of concern

Does the work involve any of these experiments of concern:

- | No | Yes |
|--------------------------|--|
| ☐ | ☐ Demonstrate how to render a vaccine ineffective |
| ☐ | ☐ Confer resistance to therapeutically useful antibiotics or antiviral agents |
| ☐ | ☐ Enhance the virulence of a pathogen or render a nonpathogen virulent |
| ☐ | ☐ Increase transmissibility of a pathogen |
| ☐ | ☐ Alter the host range of a pathogen |
| ☐ | ☐ Enable evasion of diagnostic/detection modalities |
| ☐ | ☐ Enable the weaponization of a biological agent or toxin |
| ☐ | ☐ Any other potentially harmful combination of experiments and agents |

ChIP-seq

Data deposition

- ☐ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links

May remain private before publication.

For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.

Files in database submission

Provide a list of all files available in the database submission.

Genome browser session

(e.g. [UCSC](#))

Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.

Methodology

Replicates

Describe the experimental replicates, specifying number, type and replicate agreement.

Sequencing depth

Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.

Antibodies

Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.

Peak calling parameters

Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.

Data quality

Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.

Software

Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.

Flow Cytometry

Plots

Confirm that:

- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.

Instrument

Identify the instrument used for data collection, specifying make and model number.

Software

Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.

Cell population abundance

Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.

Gating strategy

Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.

- ☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type

Indicate task or resting state; event-related or block design.

Design specifications

Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.

Behavioral performance measures

State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).

Acquisition

Imaging type(s)	<i>Specify: functional, structural, diffusion, perfusion.</i>
Field strength	<i>Specify in Tesla</i>
Sequence & imaging parameters	<i>Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.</i>
Area of acquisition	<i>State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.</i>
Diffusion MRI	☐ Used ☐ Not used

Preprocessing

Preprocessing software	<i>Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).</i>
Normalization	<i>If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.</i>
Normalization template	<i>Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.</i>
Noise and artifact removal	<i>Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).</i>
Volume censoring	<i>Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.</i>

Statistical modeling & inference

Model type and settings	<i>Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).</i>
Effect(s) tested	<i>Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.</i>
Specify type of analysis:	☐ Whole brain ☐ ROI-based ☐ Both
Statistic type for inference (See Eklund et al. 2016)	<i>Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.</i>
Correction	<i>Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).</i>

Models & analysis

n/a	Involved in the study
☐	☐ Functional and/or effective connectivity
☐	☐ Graph analysis
☐	☐ Multivariate modeling or predictive analysis
Functional and/or effective connectivity	<i>Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).</i>
Graph analysis	<i>Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).</i>
Multivariate modeling and predictive analysis	<i>Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.</i>